Project: Worked examples
Title: Thin cylindrical shell under internal pressure
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Minimum shell thickness of a thin cylindrical pressure vessel under internal pressure for circumferential and longitudinal stress, the governing required thickness with corrosion allowance and design factor, the hoop stress at the corroded thickness provided, and the provided thickness against the requirement.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Thickness needed for the circumferential stress | $t_c = P \cdot R/(S \cdot E - 0.6 \cdot P)$ |
| Thickness needed for the longitudinal stress | $t_l = P \cdot R/(2 \cdot S \cdot E + 0.4 \cdot P)$ |
| Required thickness, with the design factor and corrosion allowance | $t_{req} = n \cdot \max(t_c, t_l) + c$ |
| Hoop stress in the plate provided, after corrosion | $\sigma_\theta = P \cdot (R + 0.6 \cdot t)/(E \cdot t)$, at $t = t_p - c$ |
| Range of validity of the thin-wall formulae | $t_c \le R/2$ and $P \le 0.385 \cdot S \cdot E$ |

Basis: Thickness of shells under internal pressure, circumferential stress[1](UG-27(c)(1)); Thickness of shells under internal pressure, longitudinal stress[1](UG-27(c)(2)).

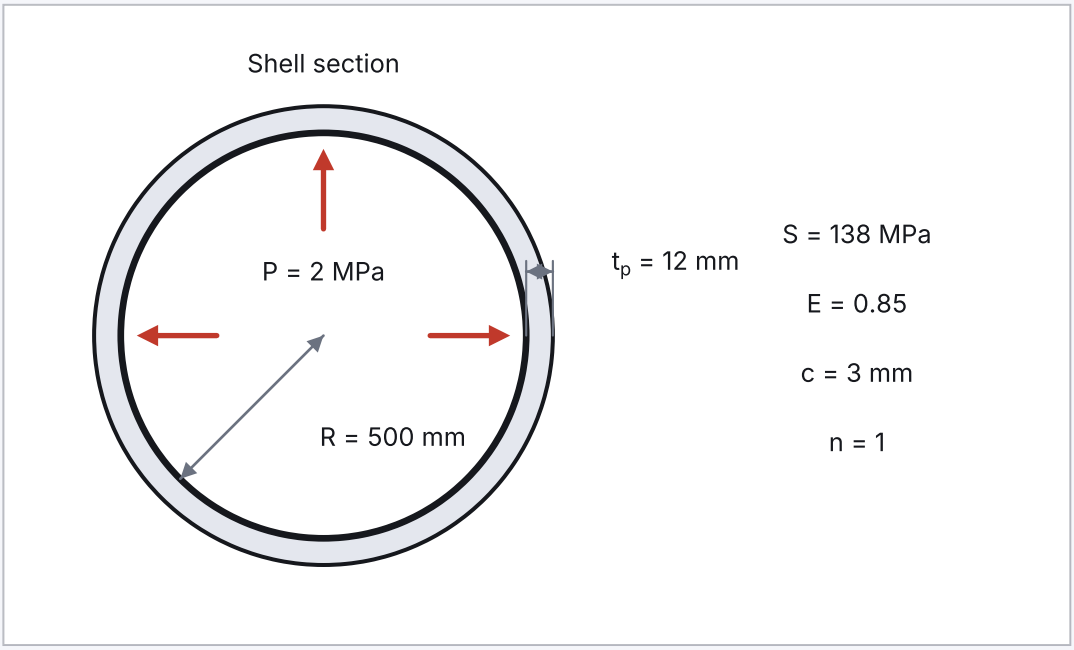



Figure 1 — A thin-walled cylinder in section under internal pressure: inside radius R, shell thickness $t_p$ including the corrosion allowance c. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| P | Internal design pressure | 2 | MPa | spec: LOAD-SCHEDULE-01 | 0.01 … 100 MPa |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| R | Inside radius (corroded) | 500 | mm | drawing: DRG-PV-601 rev A | 5 … 10000 mm |
| S | Allowable stress at design temperature | 138 | MPa | spec: SPEC-MECH-03 design criteria | 10 … 1000 MPa |
| $E_j$ | Joint efficiency | 0.85 | | spec: SPEC-MECH-03 design criteria | 0.45 … 1 |
| n | Design factor on the computed thickness (legacy; 1 for none) | 1 | | spec: SPEC-MECH-03 design criteria | 1 … 5 |
| c | Corrosion allowance | 3 | mm | spec: SPEC-MECH-03 design criteria | 0 … 25 mm |
| $t_p$ | Shell thickness provided (nominal, before corrosion) | 12 | mm | drawing: DRG-PV-601 rev A | 0.5 … 500 mm |

## 3. Assumptions

A1. Cylindrical shell away from heads, nozzles and supports; membrane stresses only. (UG-27 scope.)
A2. The joint efficiency reduces the allowable stress uniformly. (Code-form simplification carried from the legacy module.)

## 4. Working

[8] $t_c = P * R / (S * E_j - 0.6 * P)$ [1](VIII-1 UG-27(c)(1))
= 2 MPa × 500 mm / (138 MPa × 0.85 − 0.6 × 2 MPa)
= 8.6133 mm
Thickness for circumferential stress

[9] $t_l = P * R / (2 * S * E_j + 0.4 * P)$ [1](VIII-1 UG-27(c)(2))
= 2 MPa × 500 mm / (2 × 138 MPa × 0.85 + 0.4 × 2 MPa)
= 4.2481 mm
Thickness for longitudinal stress

[10] $t_{req} = n * \max(t_c, t_l) + c$
= 1 × max(8.6133 mm, 4.2481 mm) + 3 mm
= 11.613 mm
Required nominal thickness (design factor and corrosion allowance included)

[11] $t_{corr} = t_p - c$
= 12 mm − 3 mm
= 9 mm
Corroded thickness provided

[12] $\sigma_h = P * (R + 0.6 * t_{corr}) / (E_j * t_{corr})$ [1](VIII-1 UG-27(c)(1))
= 2 MPa × (500 mm + 0.6 × 9 mm) / (0.85 × 9 mm)
= 132.13 MPa
Hoop stress at the corroded thickness provided

[13] $u = t_{req} / t_p$
= 11.613 mm / 12 mm
= 0.96777
Required over provided thickness

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $t_c$ | 8.613 mm | Thickness, circumferential stress |
| $t_l$ | 4.248 mm | Thickness, longitudinal stress |
| $t_{req}$ | 11.61 mm | Required nominal thickness |
| $\sigma_h$ | 132.1 MPa | Hoop stress at the corroded thickness |
| u | 0.97 | Thickness utilisation (governing) |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Provided thickness against required | u = 0.97 | u ≤ 1 | 0.97 | [1](VIII-1 UG-27) |

## 7. Applicability

None.

## 8. Verification

Template mech.thin-vessel@1.0.1 Content hash 89ebf5d7f4e4e4fb Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. American Society of Mechanical Engineers. Boiler and Pressure Vessel Code, Section VIII, Division 1: Rules for Construction of Pressure Vessels. ASME.